



Level 3 Certificate MATHEMATICAL STUDIES 1350/2C

Paper 2C Graphical Techniques

Mark scheme

June 2025

Version: 1.0 Final



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

No student should be disadvantaged on the basis of their gender identity and/or how they refer to the gender identity of others in their exam responses.

A consistent use of 'they/them' as a singular and pronouns beyond 'she/her' or 'he/him' will be credited in exam responses in line with existing mark scheme criteria.

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Glossary for Mark Schemes

Mathematical Studies examinations are marked in such a way as to award positive achievement wherever possible. Thus, for Mathematical Studies papers, marks are awarded under various categories.

If a student uses a method which is not explicitly covered by the mark scheme the same principles of marking should be applied. Credit should be given to any valid methods. Examiners should seek advice from their senior examiner if in any doubt.

M	Method marks are awarded for a correct method which could lead to a correct answer.
A	Accuracy marks are awarded when following on from a correct method. It is not necessary to always see the method. This can be implied.
B	Marks awarded independent of method.
ft	Follow through marks. Marks awarded for correct working following a mistake in an earlier step.
SC	Special case. Marks awarded for a common misinterpretation which has some mathematical worth.
M dep	A method mark dependent on a previous method mark being awarded.
B dep	A mark that can only be awarded if a previous independent mark has been awarded.
oe	Or equivalent. Accept answers that are equivalent. eg accept 0.5 as well as $\frac{1}{2}$
[a, b]	Accept values between a and b inclusive.
[a, b)	Accept values $a \leq \text{value} < b$
3.14 ...	Accept answers which begin 3.14 eg 3.14, 3.142, 3.1416
Use of brackets	It is not necessary to see the bracketed work to award the marks.

Examiners should consistently apply the following principles

Diagrams

Diagrams that have working on them should be treated like normal responses. If a diagram has been written on but the correct response is within the answer space, the work within the answer space should be marked. Working on diagrams that contradicts work within the answer space is not to be considered as choice but as working, and is not, therefore, penalised.

Responses which appear to come from incorrect methods

Whenever there is doubt as to whether a student has used an incorrect method to obtain an answer, as a general principle, the benefit of doubt must be given to the student. In cases where there is no doubt that the answer has come from incorrect working then the student should be penalised.

Questions which ask students to show working

Instructions on marking will be given but usually marks are not awarded to students who show no working.

Questions which do not ask students to show working

As a general principle, a correct response is awarded full marks.

Misread or miscopy

Students often copy values from a question incorrectly. If the examiner thinks that the student has made a genuine misread, then only the accuracy marks (A or B marks), up to a maximum of 2 marks are penalised. The method marks can still be awarded.

Further work

Once the correct answer has been seen, further working may be ignored unless it goes on to contradict the correct answer.

Choice

When a choice of answers and/or methods is given, mark each attempt. If both methods are valid then M marks can be awarded but any incorrect answer or method would result in marks being lost.

Work not replaced

Erased or crossed out work that is still legible should be marked.

Work replaced

Erased or crossed out work that has been replaced is not awarded marks.

Premature approximation

Rounding off too early can lead to inaccuracy in the final answer. This should be penalised by 1 mark unless instructed otherwise.

Continental notation

Accept a comma used instead of a decimal point (for example, in measurements or currency), provided that it is clear to the examiner that the student intended it to be a decimal point.

Q	Answer	Mark	Comments
1 (a)	$\frac{2}{5}$	B1	

Q	Answer	Mark	Comments
1 (b)	6 m 23 s – 2 m 18 s or 4 m 5 s or 383 – 138 or 245(s) or 6.38(3...) – 2.3 or 4.08(3...) or 6 m 23 s – 3 m 44 s or 2 m 39 s or 383 – 224 or 159(s) or 6.38(3...) – 3.73(3...) or 2.65	M1	oe range for Europe or Africa
	4 m 5 s and 2 m 39 s and No or 245(s) and 159(s) and No or 4.08 (m) and 2.65 (m) and No	A1	oe must compare consistent units
	Additional Guidance		
	Condone other representation of time for method and accuracy eg 4:05 or 4.05 or 405		
	Ignore further working for comparison eg doubling the range for Africa		
	No may be implied eg Jaydon is wrong		
	6.23 – 2.18 = 4.05 and 6.23 – 3.44 = 2.79 and No		M1 A0
	623 – 218 = 405 and 623 – 344 = 279 and No		M1 A0

Q	Answer	Mark	Comments
1 (c)	Valid explanation eg He has included the 2026 eclipse (2 m 18 s) twice or He has only worked out the median/ used the times for North America (2 m 18 s and 2 m 36 s) or He did not include 2027 eclipse (in Europe, 6 m 23 s) or He found the value between the first and second value from the three values or He worked out the mean for North America (2 m 18 s and 2 m 36 s)	B1	oe
	2 m 36 s	B1	correct median must be clearly stated or on the answer line
	Additional Guidance		
	He should have just found the median of 2 m 18 s, 2 m 36 s, 6 m 23 s and correct median is 2 m 36 s		B0 B1
	He worked out the mean (not the median)		B0
	He found the median/mean of 2 m 18 s and 2 m 36 s (not all three values)		B1

Q	Answer	Mark	Comments
2 (a)	Alternative method 1: broadcasting as a proportion of total revenue		
	294 ÷ 731 or 0.4(0...) or 74 ÷ 190 or 0.38(9...) or 0.39	M1	condone reciprocals
	0.4(0...) and 0.38(9..) or 0.39 and yes or 2.48(6...) or 2.49 or 2.5 and 2.56(7...) or 2.57 or 2.6 and yes	A2	oe percentages A1 0.4(0...) and 0.38(9...) or 0.39 or 2.48(6...) or 2.49 or 2.5 and 2.56(7...) or 2.57 or 2.6
	Alternative method 2: recalculating AFC Ajax proportion		
	294 ÷ 731 (× 190)	M1	calculating AFC Ajax broadcast revenue based on Manchester City proportion
	76.(4...) (€million) and yes	A2	A1 76.(4...) (€million)
	Alternative method 3: recalculating Manchester City proportion		
	74 ÷ 190 (× 731)	M1	calculating Manchester City broadcast revenue based on AFC Ajax proportion
	284.7(0...) or 285 (€million) and yes	A2	A1 284.7(0...) or 285 (€million)

The mark scheme for Question 2 (a) continues on the next page

Q	Answer	Mark	Comments
2 (a) cont'd	Alternative method 4: Scale factor of broadcast compared to scale factor of totals		
	294 ÷ 74 or 3.9(7....) or 731 ÷ 190 or 3.8(4...)	M1	condone reciprocals
	3.9(7...) or 4 and 3.8(4...) and yes or 0.25(1...) or 0.3 and 0.25(9...) or 0.3 and yes	A1 A2	A1 3.9(7...) or 4 and 3.8(4...) or 0.25(1...) or 0.3 and 0.25(9...) or 0.3
	Additional Guidance		
	Other comparable proportions may be used eg 294 ÷ 437 = 0.67 and 74 ÷ 116 = 0.64 and yes		
	Conclusion should support statement		
	Allow rounding or truncation if method seen		
	Comparison of non-broadcast revenue		M0 A0

Q	Answer	Mark	Comments
2 (b)	Alternative method 1		
	AC Milan	B1	
	$104 \div 33$ or 3.1(51..) or 3.2 or $133 \div 33$ or 4.(03)	M1	
	3.1(51..) or 3.2 and 4.(03)	A1	matchday unit value of 1 implied by these values SC2 Real Madrid and 3.3(40...) and 3.5(16...)
	Alternative method 2		
	AC Milan	B1	
	33×3 or 99 or 33×4 or 132	M1	
	99 and 132	A1	SC2 Real Madrid and 273 and 364
	Alternative method 3		
	AC Milan	B1	
	$270 \div 8$ or $(104 + 133 + 33) \div 8$ or 33.75 or 33.8 or 34	M1	
	[101, 102] and [135,136] and [33.75,34] or [3.05, 3.09] and [3.9, 3.95] and [0.97, 0.98]	A1	SC2 Real Madrid and 268.125 and 357.5 and 89.375 or Real Madrid and 267.75 and 357 and 89.25

The mark scheme for Question 2 (b) continues on the next page

	Answer	Mark	Comments
2 (b) cont'd	Alternative method 4		
	AC Milan	B1	
	$104 \div 3$ or $34.(6\dots)$ or $133 \div 4$ or $33.(25)$	M1	
	$34.(6\dots)$ and $33.(25)$	A1	SC2 Real Madrid and $101.(3\dots)$ and 80
	Alternative method 5		
	AC Milan	B1	
	$104 - 2 \times (133-104)$ or $104 - 2 \times 29$	M1	
	46 with M1 scored	A1	SC2 Real Madrid and $304 - 2 \times (320-304) = 288$
	Alternative method 6		
	AC Milan	B1	
	$104 \div 270$ or $0.38(5\dots)$ or 0.39 or $133 \div 270$ or $0.49(\dots)$ or $33 \div 270$ or $0.12(\dots)$	M1	
	$[3.16, 3.2]$ and $[4, 4.11]$ and 1 or $0.38(5\dots)$ or 0.39 and 0.375 and $0.49(\dots)$ and 0.5 and $0.12(\dots)$ and 0.125	A1	SC2 Real Madrid and $[3.38, 3.52]$ and $[3.2, 3.4]$ or Real Madrid and $0.44(5\dots)$ or 0.45 and $0.42(5\dots)$ or 0.43 and $0.12(7\dots)$ or 0.13 and 0.375 and 0.5 and 0.125

The mark scheme for Question 2 (b) continues on the next page

2 (b) cont'd	Additional Guidance	
	Allow any equivalent method eg $\frac{4}{3} \times 104 = 138.7$ and $\frac{1}{3} \times 104 = 34.7$	

Q	Answer	Mark	Comments
2 (c)	three values added in domains [303.5, 304.5) + [319.5, 320.5) + [90.5, 91.5) that give answer in range [713.5, 714.5)	B1	
	Additional Guidance		
	Final answer not required but if seen must be correct or shown as 714		
	303.5 + 319.5 + 90.5		B1
	304.5 + 319.5 + 90.5		B0
	303.4 + 320 + 91		B0

Q	Answer	Mark	Comments
2 (d)	$270 \div 1.16$ or 233 or 232.(7...)	M1	oe
	$144 \div 0.94$ or 153.(19...) or 153.2	M1	oe
	80 with M2 scored	A1	condone values stated in millions
	Additional Guidance		
	153 from incorrect method eg $144 \times 1.06 = 152.64 = 153$		M0
	153 and 233 and 80 with no incorrect method		M1 M1 A1

Q	Answer	Mark	Comments	
2 (e)	Any two valid criticisms from distinct categories	B2	B1 one valid criticism allow improvements to imply criticism two criticisms may be in one response ignore any extra criticisms	
	Shape Pyramid / 3D Does not start at zero Shape is above and below the horizontal axis Shape does not align with scale Shapes start from different points			
	Axis label No <i>y</i> -axis label No <i>x</i> -axis label Angle of labels No units eg currency or millions			
	Scale / Gridlines Gridlines only every 50 units Negatives signs are on the gridlines The gaps on the y-scale are too large The vertical axis only extends to 50 (so the 49 label is outside the graph area)			
	Additional Guidance			
	-25 is too close to -50 on the scale (Shape category)			B1
	Should have used rectangles (Shape category)			B1
	More gridlines (Gridlines category)			B1
	Should extend the vertical axis beyond 50 (Scale / Gridlines category)			B1
	Inaccurate			B0
No vertical gridlines	B0			
No gridlines	B0			

Q	Answer	Mark	Comments
2 (f) (i)	$(49 + 21 + 13 + 9 - 24 - 25 - 69 - 369) \div 8$ or $(-) 395 \div 8$	M1	
	– 49.375 with M1 scored or loss of 49.375 with M1 scored	A1	allow rounding or truncating SC1 $- 400 \div 8 = -50$ or $(-)49.3(75)$ or $(-)49.38$ or $(-)49.4$ without working
	Additional Guidance		
	Ignore units		
	Calculating the median or other measures does not support the statement		
	$49 + 21 + 13 + 9 - 24 - 25 - 69 - 369 \div 8$		M0 unless recovered
	SC1 awarded for rounding $(-)395$ to $(-)400$ prior to division		
	The mean is $(-)49$ without working		M0 A0

Q	Answer	Mark	Comments
2 (f) (ii)	One valid reason eg The data is old / only from one year/season The clubs are not representative of current European champions The clubs are not representative of champions from other European leagues Only used a small sample of clubs PSG’s loss has skewed the mean average / PSG is an outlier The median average is a loss of €7.5m Without PSG the mean loss is only €3.7 m Data affected by Covid	B1	
	Additional Guidance		
	The majority of clubs do not make a loss of more than €50m		B1
	Half of the clubs made a profit		B0
	Only represents clubs who win their league		B0
	There was an outlier		B0
	None of the clubs are close to losing the mean amount		B0

Q	Answer	Mark	Comments
3 (a)	Third box indicated	B1	
	Additional Guidance		

Q	Answer	Mark	Comments
3 (b)	Alternative method 1		
	Horizontal line seen from 200 cm	M1	implied by correct intersection mark or line on horizontal axis or 48
	48 (inches)	A1	implied by correct answer
	43	B1ft	ft correct TV size for their 48
	Alternative method 2		
	Vertical line seen at 43 inches and vertical line seen at 50 inches	M1	implied by correct intersection marks or lines on vertical axis
	[170, 190] (cm) and [200, 220] (cm)	A1	
	43	B1	
	Alternative method 3		
	gradient of line = [4, 4.2]	M1	
	[47.5, 48] (inches)	A1	
	43	B1ft	ft correct TV size for their [47.5, 48]
	Additional Guidance		

Q	Answer	Mark	Comments
4 (a)	$(b =) 396\,978$	B1	
	$404\,219 - 396\,978$ or 7241	M1	
	their $7241 \div 10$	M1	Their 7241 must come from $404\,219 - 396\,978$
	$(a =) 724(.1)$	A1	
	Additional Guidance		

Q	Answer	Mark	Comments
4 (b) (i)	Values of Y 16 and 18 and 20	B1	
	326 000 and 460 000 and 651 000	B2ft	ft their 16 and 18 and 20 B1ft for non-rounded answers 325 765.(7...) 460 438.(1...) 650 784.(4...) or correct answers rounded to a different amount of significant figures or two correct answers
	Additional Guidance		

Q	Answer	Mark	Comments
4 (b) (ii)	Correct linear plot for M	B1ft	ft their values from question 4(a) must be for domain $0 \leq Y \leq 20$ within the correct grid square
	their values of N correctly plotted	M1	ft their values from question 4(b)(i) implied by a curve that goes through their three coordinates within the correct grid square
	Smooth curve for N drawn	M1	must go through their coordinates even if not plotted as points
	Reading below their intersection on horizontal axis or [17, 18] with intersection seen	M1	can be implied by year consistent with their intersection
	2028 or 2029 with the first M2 awarded and an intersection seen	A1ft	ft graphs correct to their question 4(a) and question 4(b)(i)
	Additional Guidance		
	Linear coordinates need to be connected by a straight line		
	Coordinates for linear plot (0, 397) (10,404) (20, 411)		
	Coordinates for curve (16, 326) (18, 460) (20, 651)		
	An intersection reading from the horizontal line $N = 350$		M0
	Correct linear plot and correct smooth curve and 2028 or 2029		B1 M1 M1 M1 A1
	Only two coordinates for N plotted but joined correctly with curve		M0 M1

Q	Answer	Mark	Comments
4 (b) (iii)	Trend(s) may not continue or Linear model may not be true for Maltese citizens or Exponential model may not be true for non-Maltese citizens or Atypical values used (to create model)	B1	allow reasons to imply change in trend eg Covid
	Additional Guidance		
	Unlikely the trend will continue (very large values from exponential)		B1
	Extrapolation is unreliable		B1
	Unreliable after 2021		B1
	Formula may not be true		B1
	There may have been a change (since 2021) eg birthrate drop		B1
	There may be a pandemic		B1
	Does not account for deaths or people leaving the country		B0
	The values were rounded		B0
	It is an estimate		B0
	It assumes the amount(s) will increase the same each year		B0

Q	Answer	Mark	Comments
4 (b) (iv)	Constant rate for Maltese citizens and Increasing rate for non-Maltese citizens	B2	oe B1 constant rate for Maltese citizens or increasing rate for non-Maltese citizens or linear growth for Maltese citizens and exponential growth for non-Maltese citizens or the rate for non-Maltese citizens is (always) higher than the rate for Maltese citizens or The non-Maltese population increases faster than Maltese must specify correct population condone using M and N rather than Maltese and non-Maltese
	Additional Guidance		
	Do not accept linear as a direct replacement for ‘constant’		
	Non Maltese is increasing and Maltese is linear		B1
	Constantly increasing		B0

Q	Answer	Mark	Comments
5 (a)	$(a =) [1320, 1380]$	B1	implied by final answer
	$(d =) [19, 19.5]$	B1	implied by final answer
	$(t =) \frac{\text{their } d}{4} + \frac{\text{their } a}{600}$ or $[6.95, 7.2]$	M1	ft their d and their a
	their $d \div$ their t or $[2.63, 2.81]$	M1	estimated speed using Naismith time
	Their $d \div 6$ or $[19, 19.5] \div 6$ or $[3.16, 3.25]$	M1	actual speed
	$[0.35, 0.62] \text{ (km/h) with M3 scored}$	A1	difference between speeds
	His actual speed was greater than predicted and $[0.35, 0.62]$	E1ft	ft correct decision for their difference SC1 states their real speed was greater because they took less time, with no other working shown
	Additional Guidance		
	Allow answers in m/s $[2.63, 2.91] = [0.73, 0.81] \text{ m/s}$ $[3.16, 3.25] = [0.87, 0.91] \text{ m/s}$ $[0.35, 0.62] = [0.09, 0.18] \text{ m/s}$		

Q	Answer	Mark	Comments
5 (b)	$6000 \div [680, 720]$ or $6 \div [0.68, 0.72]$ or $[8.3, 8.8]$	M2	M1 correct calculation not compensating for conversion of km to m
	$[8.3, 8.8]$ rounded to 2 significant figures	A1	correct for their $[680, 720]$ rounded to 2 significant figures without contradiction
	Additional Guidance		
	$(6000 \div 700 =) 8.57$		M2 A0
	$(6000 \div 700 =) 8.6$		M2 A1

Q	Answer	Mark	Comments
6 (a)	0	B1	
	Gradient at maximum point is zero or Tangent to maximum velocity is horizontal or It is the turning point / stationary point of the graph	E1	may refer to drawing of a horizontal tangent
	Additional Guidance		
	m = 0 at maximum point		E1
	Line/curve is flat at maximum point		E1
	Gradient or m = 0		E0
	Line/curve is flat		E0
	They are travelling fastest so can not accelerate any more		E0

Q	Answer	Mark	Comments
6 (b)	Tangent drawn to curve at $t = 4$	M1	tangents need to have a gradient in range [60, 100]
	Gradient calculated using $\frac{\text{difference in velocity}}{\text{difference in time}}$	M1	may be from an invalid tangent or from two points on the curve values used must be correct for their tangent or triangle
	[60, 100] with tangent drawn	A1	
	Additional Guidance		
	True value is 81.7		

Q	Answer	Mark	Comments
7 (a)	$A = 250$	B1	may be seen or implied at any stage when using the equation
	$\frac{S}{A} = e^{kH}$ or $8880 = e^{10k}$	M1	oe
	$kH = \ln \frac{S}{A}$ or $k = \frac{\ln \frac{S}{A}}{H}$ or $10k = \ln(8880)$	M1	oe award any correct application of ln
	$k = [0.9, 0.91]$	A1ft	ft their A
	Additional Guidance		
	$Ae^{0k} = 250$		B0

Q	Answer	Mark	Comments	
7 (b)	The Scoville units and heat rating will change by different factors or not proportional or Scoville units is exponential and heat rating is linear or Would need to use the (exponential or logarithm) formula and the heat rating will be less than 13 or ($H =$) [10.3, 10.4] or formula predicts $S =$ [30 000 000, 35 000 000] when $H = 13$ not 2 886 000	B2	B1 the Scoville units and heat rating will change by different factors or not proportional or Scoville units is exponential and heat rating is linear or would need to use the (exponential or logarithm) formula or the heat rating will be less than 13 and ($H =$) [10.3, 10.4] or formula predicts $S =$ [30 000 000, 35 000 000] when $H = 13$ not 2 886 000	
	Additional Guidance			
	Use of formula can be implied by explaining the calculation they would need to use eg they should have done $2220000 \times 1.3 = 250e^{0.9H}$			